



USER GUIDE

HIAS

Hall Invigilator Allocation System

St. Xavier's Catholic College of Engineering (Autonomous), Nagercoil

Version **1.0.0** · Windows desktop application · Works fully offline

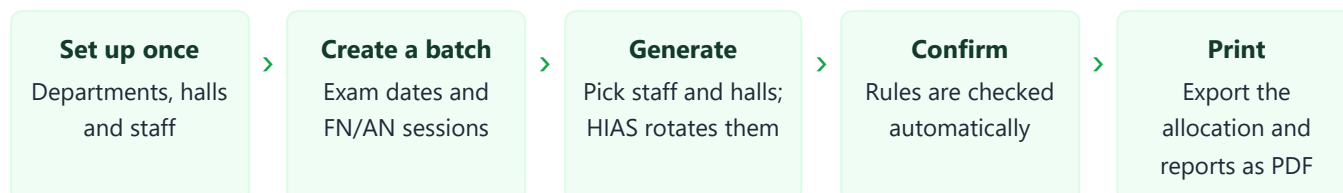
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Contents

- 1 What HIAS does
- 2 Installing HIAS
- 3 A tour of the screen
- 4 First-time setup: college, departments, halls and staff
- 5 Creating an allocation batch
- 6 Allocating invigilators to a session
- 7 Reports
- 8 Allocation history
- 9 Backup and restore (important)
- 10 Restarting the rotation
- 11 Troubleshooting and FAQ

1. What HIAS does

HIAS assigns invigilators (staff) to examination halls for every exam session, and rotates those assignments fairly. It remembers who has already been in which hall, so nobody is sent to the same hall again until they have had a turn in every other hall of the rotation.



- **Fully offline.** HIAS needs no internet connection and no server. Everything is stored on the computer it is installed on.
- **No login.** The app opens straight to the dashboard. Keep it on a computer that only authorised people use.
- **Automatic rule checks.** Before an allocation is confirmed, HIAS checks that no hall or person is double-booked, nobody repeats a hall too early, and inactive staff or halls are not used.

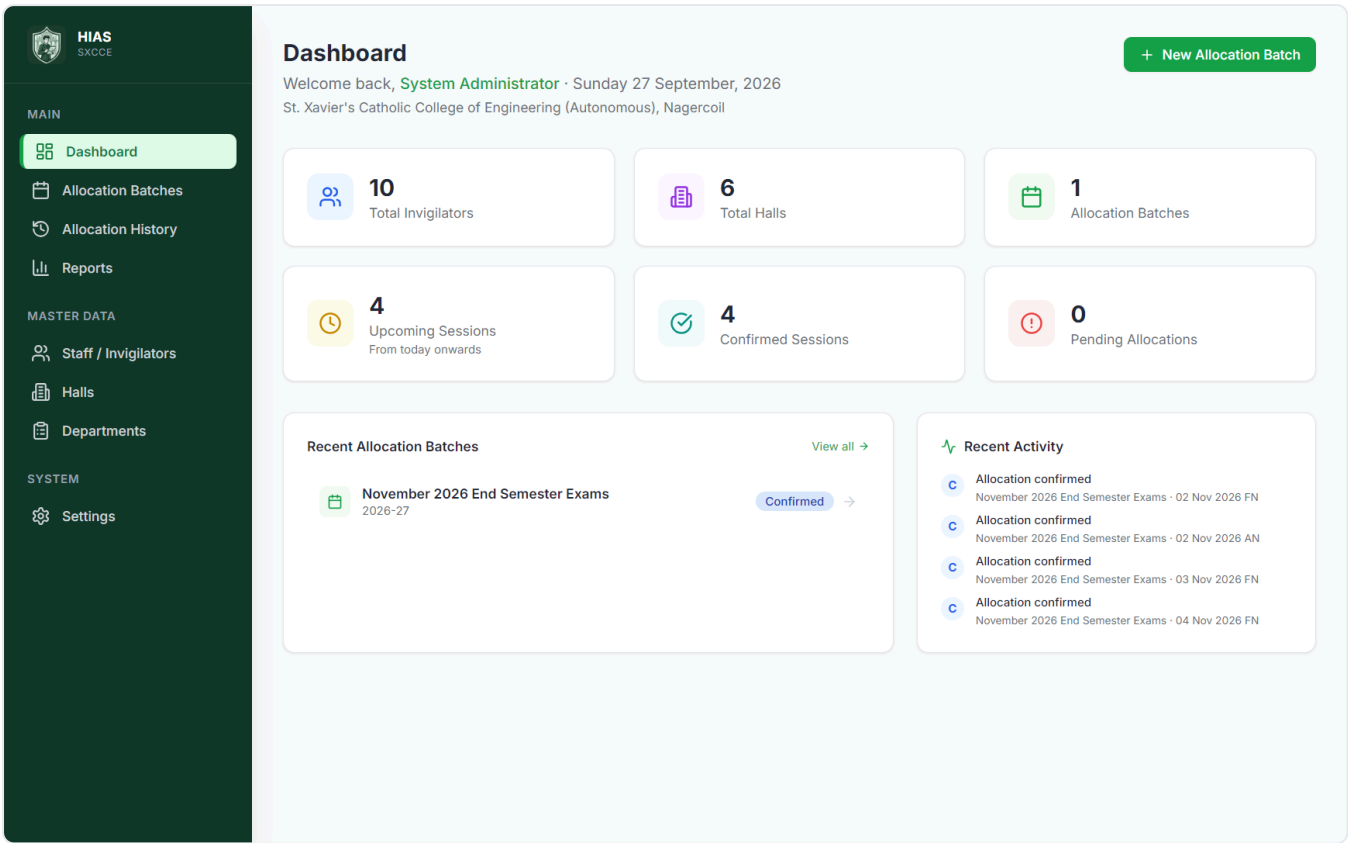
2. Installing HIAS

- 1 Copy `HIAS-Setup-1.0.0.exe` (about 73 MB) to the computer and double-click it.
- 2 If Windows shows "*Windows protected your PC*", click **More info** and then **Run anyway**. This appears because the installer is not code-signed; it is safe.
- 3 When Windows asks "*Do you want to allow this app to make changes to your device?*", click **Yes**. HIAS installs for everyone who uses the computer, which needs administrator permission once.
- 4 Choose the installation folder (the default is fine) and click **Install**.
- 5 Leave **Run HIAS** ticked and click **Finish**. Next time, open HIAS from the **HIAS** shortcut on the desktop or in the Start menu.

Installing a newer version over an older one upgrades it and keeps all data. To remove HIAS, use Windows **Settings** → **Apps** → **Installed apps** → **HIAS - Hall Invigilator Allocation System** → **Uninstall**.

Where is my data? All data is kept in one file on this computer: `%APPDATA%\eias\eias.db`. Updating or reinstalling HIAS keeps this file. Uninstalling the app does not delete it either. See section 9 for backups.

3. A tour of the screen



The dashboard. The sidebar on the left is always available. (All names in this guide are sample data.)

Sidebar item	What it is for
Dashboard	Totals at a glance: invigilators, halls, batches, upcoming and confirmed sessions, and sessions still pending allocation. Click a card to jump to it.
Allocation Batches	Create an exam batch (for example "November 2026 End Semester Exams") and open it to allocate each session.
Allocation History	Search every confirmed allocation by batch, session, staff member or hall.
Reports	Staff-wise, date-wise, timetable and audit reports, each exportable as PDF.
Staff / Halls / Departments	The master data HIAS allocates from.
Settings	College name, default session timings, backup and restore, and rotation restart.

A **Back** link at the top of each page returns to the previous screen.

4. First-time setup

Do these steps once, in this order. You can come back and change anything later.

4.1 College profile and session timings

Open **Settings** → **College Profile**. The **College Full Name** is printed at the top of every PDF; the **Short Name** appears under the HIAS logo in the sidebar. Click **Save**.

Under **Settings** → **Session Timings**, set the usual reporting, start and end times for the Forenoon (FN) and Afternoon (AN) sessions. New batches start with these times; you can still change them per session.

4.2 Departments

Open **Departments** and add each department with a short **Code** (for example **CSE**) and a full **Name**. Reports use the short code.

4.3 Halls

Open **Halls** → **Add Hall**. Only the **Hall Code** (for example **H101**) is required; if you leave **Hall Name** empty, the code is used as the name. Capacity and Block are optional. Untick **Active** for a hall that is temporarily unavailable; inactive halls are never allocated.

ORDER	HALL CODE	NAME	BLOCK	CAPACITY	STATUS	
#1	H101	Main Block Room 101	Main Block	30	Active	
#2	H102	Main Block Room 102	Main Block	30	Active	
#3	H103	Main Block Room 103	Main Block	30	Active	
#4	H104	Main Block Room 104	Main Block	30	Active	
#5	H105	Main Block Room 105	Main Block	30	Active	
#6	H106	Main Block Room 106	Main Block	30	Active	

Halls. The pencil edits a hall; the bin removes a hall that has never been allocated (for a used hall, untick Active instead).

4.4 Staff (invigilators)

Add staff one at a time with **Add Staff**, or many at once with **Import Excel**. The file **HIAS-Staff-Import-Template.xlsx**, supplied with HIAS, has the right headings ready. For an import, the first row of the sheet must contain these column headings:

Column	Required?	Notes
Staff ID	Yes	Must be unique. Rows whose Staff ID already exists are skipped.
Name	Yes	Full name of the staff member.
Department	Yes	The department's code or full name; it must already exist in Departments.
Designation, Email	No	Optional details.

After the import, HIAS shows how many staff were added and lists each skipped row with the reason (for example *"Row 5: Department CIVIL does not match any existing department"*), so you can correct the sheet and import it again.

The screenshot shows the HIAS Master Data interface. On the left is a dark green sidebar with a menu. The main content area is light blue and titled 'Master Data' with the subtitle 'Manage staff, halls, and departments'. Below the title are three tabs: 'Staff / Invigilators' (selected), 'Halls', and 'Departments'. At the top right of the main area are two buttons: 'Import Excel' and '+ Add Staff'. Below the tabs, it says '10 staff members'. A table lists the staff members with columns: STAFF ID, NAME, DEPARTMENT, DESIGNATION, and STATUS. Each row has edit and delete icons at the end.

STAFF ID	NAME	DEPARTMENT	DESIGNATION	STATUS
STF001	Anitha R	Computer Science and Engineering	Assistant Professor	Active
STF004	Dinesh K	Computer Science and Engineering	Assistant Professor	Active
STF007	Gokul P	Computer Science and Engineering	Assistant Professor	Active
STF101	Imran K	Computer Science and Engineering	Associate Professor	Active
STF002	Bala S	Electronics and Communication Engineering	Assistant Professor	Active
STF005	Esther J	Electronics and Communication Engineering	Assistant Professor	Active
STF008	Hema V	Electronics and Communication Engineering	Assistant Professor	Active
STF102	Janani L	Electronics and Communication Engineering	Assistant Professor	Active
STF003	Chitra M	Mechanical Engineering	Assistant Professor	Active
STF006	Faizal A	Mechanical Engineering	Assistant Professor	Active

Staff / Invigilators, with Import Excel and Add Staff at the top right.

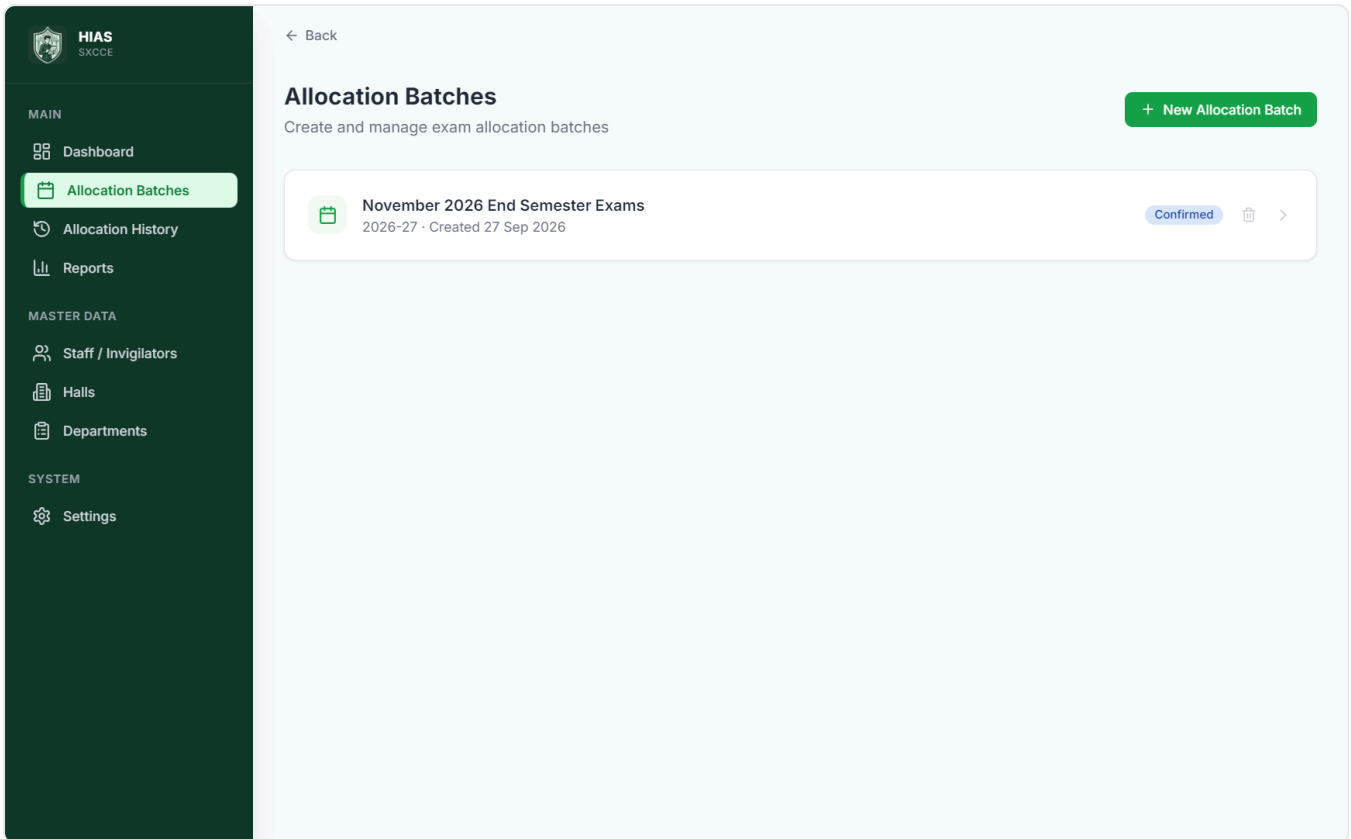
Deactivate or delete? The bin icon first **deactivates** a staff member: they stay in past records but are no longer allocated. An inactive staff member with no allocation history can then be **deleted permanently**; HIAS asks you to type `DELETE` to confirm.

5. Creating an allocation batch

A **batch** is one examination series, such as an end-semester or internal assessment exam. Open **Allocation Batches** → **New Allocation Batch**. The wizard has three steps:

- 1 Batch details.** Enter the batch name, the academic year (for example `2026-27`) and the number of exam days.

- 2 **Pick the dates** on the calendar. Sundays and public holidays are highlighted so you do not pick them by mistake.
- 3 **Sessions and timings.** For each date, choose Forenoon (FN), Afternoon (AN) or both. Timings come from Settings; expand a date to change its times.



Allocation Batches. A batch shows **Draft** until every session in it is confirmed, then **Confirmed**.

To delete a whole batch, click its bin icon and type **DELETE**. This removes all its sessions, allocations and rotation records and cannot be undone.

6. Allocating invigilators to a session

Open a batch to reach its **Allocation Workspace**. Each session appears as a tab (for example *02 Nov 2026 FN*).

- 1 Select a session tab and click **Select Staff & Generate**.
- 2 Tick the staff who are on duty and the halls in use for this session. **The number of staff must equal the number of halls**; HIAS tells you if they differ.
- 3 Click **Generate Allocation**. HIAS assigns everyone at once so that each person moves on to a hall they have not had recently, spreading halls evenly over time.
- 4 Review the list. To change one assignment, click the pencil on that row, choose a hall and, if you like, note the reason. HIAS only offers halls that keep the rules satisfied. **Regenerate** starts the session over.
- 5 Click **Confirm Allocation**. HIAS runs all rule checks; if something is wrong it explains what, and nothing is saved until it is fixed.

6 Click **Export PDF** to save the printable allocation sheet for this session.

MAIN

- Dashboard
- Allocation Batches
- Allocation History
- Reports

MASTER DATA

- Staff / Invigilators
- Halls
- Departments

SYSTEM

- Settings

← Back

November 2026 End Semester Exams

2026-27 · Allocation Workspace

02 Nov 2026 FN

02 Nov 2026 AN

03 Nov 2026 FN

04 Nov 2026 FN

+ Add Session

Date: Mon, 02 Nov 2026

Session: Forenoon (FN)

Report: 09:30 AM

Exam: 10:00 AM – 01:00 PM

Confirmed

Edit Schedule & Timings

Export PDF

#	STAFF ID	NAME	DEPARTMENT	ASSIGNED HALL	TYPE
1	STF001	Anitha R	CSE	H101	Auto-generated
2	STF004	Dinesh K	CSE	H104	Auto-generated
3	STF002	Bala S	ECE	H102	Auto-generated
4	STF005	Esther J	ECE	H105	Auto-generated
5	STF003	Chitra M	MECH	H103	Auto-generated
6	STF006	Faizal A	MECH	H106	Auto-generated

6 invigilators · 0 admin-edited

A confirmed session. Rows you changed by hand are marked "Admin Edited" instead of "Auto-generated".

Why confirming matters. The rotation only moves forward when a session is **confirmed**. Always confirm sessions in date order, so each session continues the rotation from the one before it.

The rules HIAS checks

Rule	Meaning
R1	A staff member does not return to a hall until they have been through the other halls in the rotation. When the people on duty change a lot, this is occasionally impossible for everyone; HIAS then keeps repeats to the minimum and shows them as yellow "Unavoidable repeat" warnings, which do not block confirmation.
R2	A hall gets only one invigilator per session.
R3	A staff member is in only one hall per session.
R4	Sessions are confirmed in rotation order.
R5 / R6	Inactive staff and inactive halls are never used.
R7	The number of staff equals the number of halls.

7. Reports

Open **Reports**, choose a report type, choose the filters, click **Generate** to preview it, then **Export PDF**. Each PDF carries the college name and the time it was generated. When you export, Windows asks where to save the file (the Downloads folder is suggested).

Report	Use it for	Filters
Staff-Wise Duty Report	Every duty of each staff member: date, session, hall and timings.	Optional: one staff member
Date-Wise Allocation Sheet	The hall-by-hall list for one session, for the notice board. The title names the date and session.	Batch and session
Complete Timetable Grid	One page showing every staff member against every session of a batch.	Batch
Rotation Audit Report	Which assignments were generated automatically and which were changed by an admin, with the reason.	Batch

MAIN

Dashboard

Allocation Batches

Allocation History

Reports

MASTER DATA

Staff / Invigilators

Halls

Departments

SYSTEM

Settings

← Back

Reports

Generate and export allocation reports with your college letterhead

Staff-Wise Duty Report
All sessions per staff member

Date-Wise Allocation Sheet
Hall-by-hall seating for posting

Complete Timetable Grid
Full Staff × Session matrix

Rotation Audit Report
Auto vs admin-edited assignments

Allocation Batch *

Session *

Generate

Export PDF

HALL	HALL NAME	CAP	STAFF ID	NAME	DEPT	DESIGNATION	REPORT	EXAM
H101	Main Block Room 101	30	STF001	Anitha R	CSE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H102	Main Block Room 102	30	STF002	Bala S	ECE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H103	Main Block Room 103	30	STF003	Chitra M	MECH	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H104	Main Block Room 104	30	STF004	Dinesh K	CSE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H105	Main Block Room 105	30	STF005	Esther J	ECE	Assistant Professor	09:30 AM	10:00 AM–01:00 PM
H106	Main Block Room 106	30	STF006	Faizal A	MECH	Assistant Professor	09:30 AM	10:00 AM–01:00 PM

6 records

Date-Wise Allocation Sheet preview, ready to export.

8. Allocation history

Allocation History lets you look up any past confirmed allocation. Filter by batch, session, staff member or hall and click **Search Records**. This is the quickest way to answer "Where was this person on that day?".

HIAS – Hall Invigilator Allocation System

Page 9 of 11

9. Backup and restore

HIAS keeps its data only on this computer. If the computer fails and there is no backup, the data cannot be recovered. Take a backup after every allocation batch and keep it on a pen drive or another computer.

To take a backup

- 1 Open **Settings** → **Backup & Restore** and click **Backup Database**.
- 2 Choose where to save the file (for example a pen drive). The file is named `hias-backup.db` by default; adding the date to the name helps.

To restore a backup

- 1 Open **Settings** → **Backup & Restore** and click **Restore Database**.
- 2 Choose the backup file and confirm. HIAS first saves a safety copy of the current data (`eias.db.bak` in the data folder), then replaces it.
- 3 HIAS restarts by itself a moment later and opens with the restored data.

The same steps move HIAS to a new computer: back up on the old computer, install HIAS on the new one, and restore.

10. Restarting the rotation

Settings → **Restart Rotation** clears the whole rotation history, so every staff member starts again from the first hall. Use it only at a genuine new start, such as the beginning of an academic year. You must type `RESTART` to confirm. Past allocations and reports are not deleted; only the rotation memory is cleared.

11. Troubleshooting and FAQ

Problem	What to do
"Staff count must equal hall count"	Select the same number of staff and halls for the session.
A staff member or hall is missing from the selection list	Check it is marked Active in Staff or Halls.
Confirm is refused with a rule message (R1–R7)	Read the message; it names the person or hall. Edit that row, or regenerate, then confirm again.
Windows says "Windows protected your PC" when installing	Click More info → Run anyway . The installer is not code-signed.
The app shows an error box at start-up	Note the message, then restore the latest backup (section 9). The message is shown because a silent failure would give no clue.
Can two computers share the same data?	No. Each installation has its own data. Use one computer for allocation, and move data with backup and restore if needed.
Does HIAS need the internet?	No. It works completely offline.